

Chunyang Luo

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Research Interests: Agent Systems | Human-Agent Interaction | LLM-based Planning & Reasoning | Multi-Robot Planning

EDUCATION

B.Eng. Candidate in Robotics Engineering, Peking University

2023 – Present

Expected Graduation: June 2027; GPA: 3.526/4.0 (85/100); Rank: 5/19 (Top 26%)

Selected Core Courses: Foundations of Machine Learning (98), Probability and Statistics (94), Introduction to Robotics (89)

RESEARCH EXPERIENCES

OpenGRAIL: Integrating LLM and Classical Planning for Multi-Robot Task Planning in Open Worlds

Sept. 2025 – Present

Supervisor: Prof. Zhongkui Li

- Developed an LLM–classical planning framework that translates natural-language instructions into executable multi-robot plans.
- Designed hierarchical task decomposition and MILP-based heterogeneous robot allocation and scheduling under capability, temporal, spatial, and collaboration constraints.
- Integrated planning with AI2-THOR for closed-loop execution and environment feedback. **Manuscript in preparation for ICRA 2027.**

Human-AI Collaboration for Long-Horizon Agent Systems

Jul. 2026 – Sept. 2026

Summer Research, Hong Kong University of Science and Technology (HKUST)

Supervisor: Prof. Xiaojuan Ma

- Investigated memory and workflow representations that preserve user intent, task constraints, and reusable knowledge across long-horizon human-agent collaboration.
- Studied memory organization, lifecycle management, conflict handling, and cross-task knowledge reuse to connect high-level user intent with agent execution.

Multi-Robot Collaborative LTL Continuous Planning for Dynamic Environments

Jul. 2025 – Present

Supervisor: Prof. Zhongkui Li

- Designed an online task decomposition and replanning framework for heterogeneous multi-robot LTL tasks under dynamic events.
- Built a heterogeneous multi-robot simulation platform (Unitree Go2, Scout) with task execution and state feedback interfaces.
- Benchmarked against static and periodic-replanning baselines on completion rate, planning time, and cumulative cost; extending to physical robots.

AWARDS & COMPETITIONS

- **Second Prize**, 22nd “Jiang Zehan Cup” Modeling and Computer Applications Competition, Peking University
- **Honorable Mention**, MCM/ICM 2026

SELECTED PROJECTS & TRAINING

RepoFlow: Repository-Level Workflow for AI Coding Agents (*Course Project, Peking University*)

Mar. 2026 – Present

Designed persistent-context and structured agent-handoff mechanisms for task continuity in long-horizon coding agents.

Alibaba DAMO Academy – Embodied Intelligence Bootcamp

Spring 2026

Built a robotic-arm visual-navigation demo integrating perception, control, and task execution.

SKILLS AND OTHERS

- **Programming & Tools:** Python, MATLAB, Linux, Git, LaTeX
- **Planning & AI:** LTL, PDDL, Classical Planning, Multi-Robot Task Allocation, LLM-based Planning & Reasoning, RL
- **Robotics:** ROS, Gazebo, RViz, AI2-THOR
- **Language:** CET-6 578, IELTS Overall 7.0